GOVERNMENT THIRUMAGAL MILLS COLLEGE GUDIYATHAM

(Affiliated by THIRUVALLUVAR UNIVERSITY)

GUDIYATHAM, VELLORE DISTRICT-632 602.

Project Title

Farm Manager System

TEAM LEADER

NAME : S.BALAMURUGAN

NMID : 07DC50B4A7AF5609087469A4E5F788C8

REGISTER NO : 31024U18003

TEAM MEMBER

NAME : M. DHARSHINI

NMID : 854F4709748CB1E96FC4755117E0A4F4

REGISTER NO : 31024U18007

TEAM MEMBER

NAME : P.KAVIYARASAN

NMID : CEF123AD26B0C398C2E4A783162D1CA1

REGISTER NO : 31024U18024

TEAM MEMBER

NAME : R POORNIMA

NMID : 29D3B03E78CE774ABFA4383CBEC2439E

REGISTER NO : 31024U18038

TEAM MEMBER

NAME : V.DHANUSH

NMID : C9890E23CC54015AA214FB61669793B3

REGISTER NO : 31024U18006

FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose:

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

Phase 1:

BRAINSTORMING & IDEATION PHASE

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrecT	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

1.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis:

- **Think & Feel:** Need accurate and timely farm data

- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets
- **Say & Do:** Enter records manually
- **Pain:** Data loss, errors, delays
- **Gain:** Automation, fast reporting



1.3 Brainstorming

Brainstormed Ideas:

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

PHASE 2: REQUIREMENT ANALYSIS PHASE

2.1 Customer Journey Map

Stage	User Action	Touchpoints	Pain Points	System Solution (FMS)
Awareness	Realizes the need for digital farm records	Agricultural officers, Peers	Manual errors, data loss	Introduction of centralized Salesforce FMS
Consideration	Evaluates Salesforce platform	Demo sessions, Admin	Doubts about ease of use	User-friendly UI and easy navigation
Adoption	Registers in the FMS	Login Page, Profile	Initial setup time	Quick onboarding with predefined fields
Daily Use	Enters crop & farm data regularly	Farm Object, Crop Object	Time-consuming manual entry	Quick data entry with validation rules
Reporting	Checks monthly production & sales	Dashboards, Reports	Takes time to calculate manually	Real-time automated Salesforce dashboards



2.2 Solution Requirements

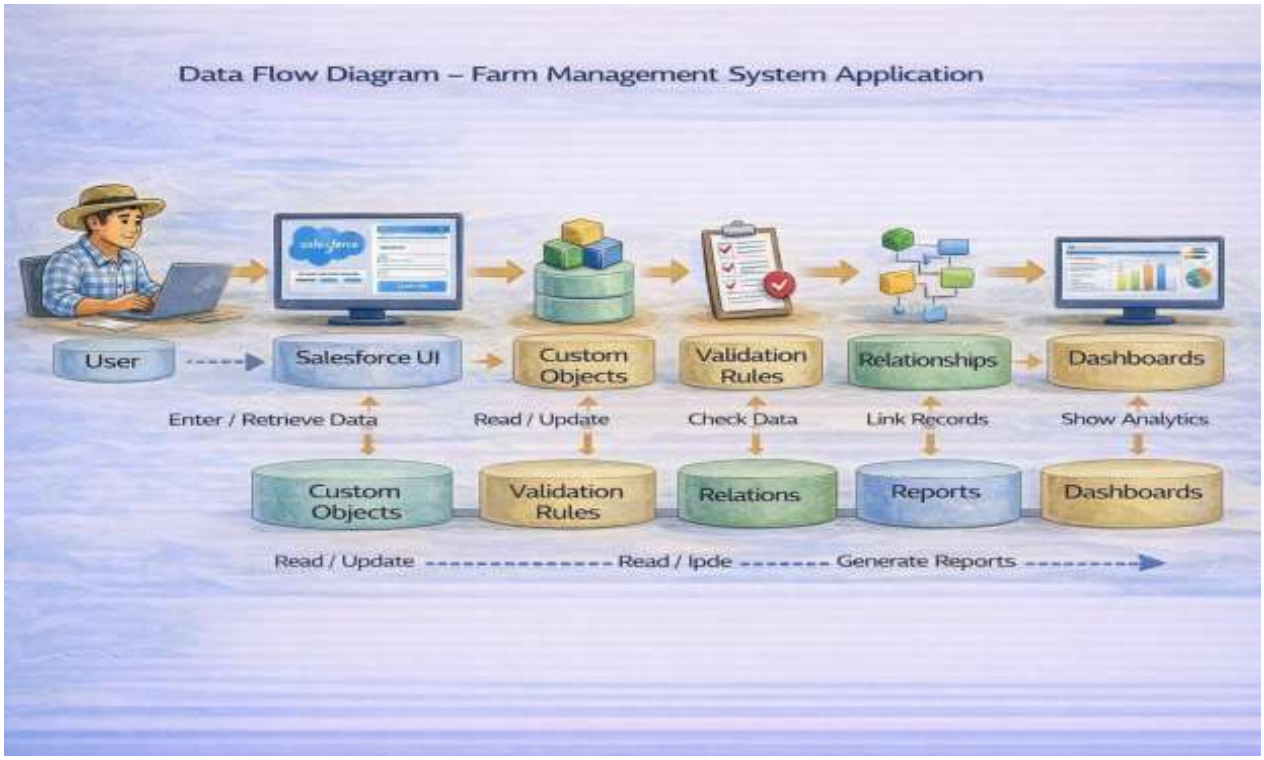
Functional Requirements (FR)

Req ID	Requirement Description	Priority
FR-01	The system must allow users to create, update, and manage Farm records.	High
FR-02	The system must allow users to track Crop cycles, quantities, and yields.	High
FR-03	The system must record and manage Farmer and Buyer profile details securely.	High
FR-04	The system must generate real-time reports and interactive dashboards to monitor farm productivity and sales.	High

Non-Functional Requirements (NFR)

Req ID	Requirement Description	Priority
NFR-01	Role-based access control must be implemented for security.	High
NFR-02	The interface should be simple, user-friendly, and easy to navigate.	Medium
NFR-03	The system must be highly responsive (updating records quickly).	High
NFR-04	Data must be stored securely on the Salesforce cloud to prevent loss.	High

2.3 Data Flow Diagram



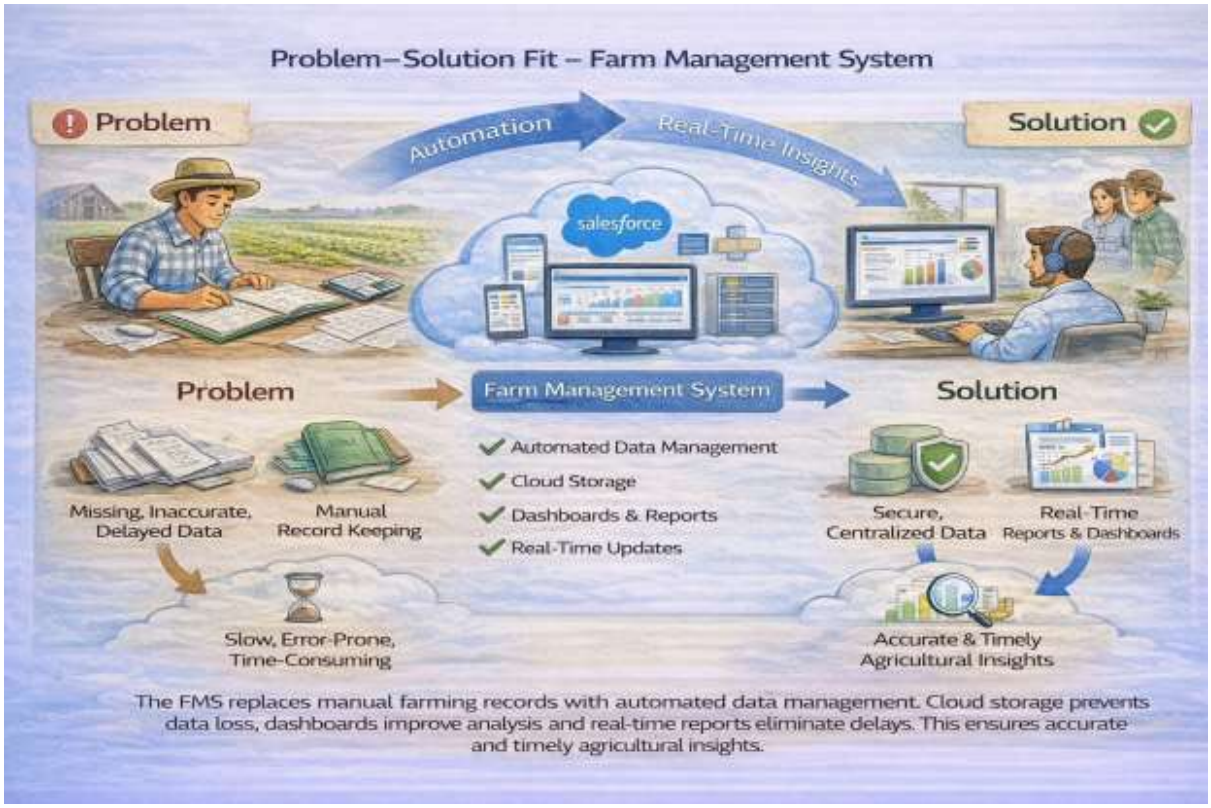
2.4 Technology Stack

Category	Technology / Platform
Cloud Platform	Salesforce CRM (Developer Edition)
User Interface (UI)	Salesforce Lightning Experience
Database	Salesforce Standard & Custom Objects
Analytics	Salesforce Reports and Dashboards

PHASE 3: PROJECT DESIGN PHASE

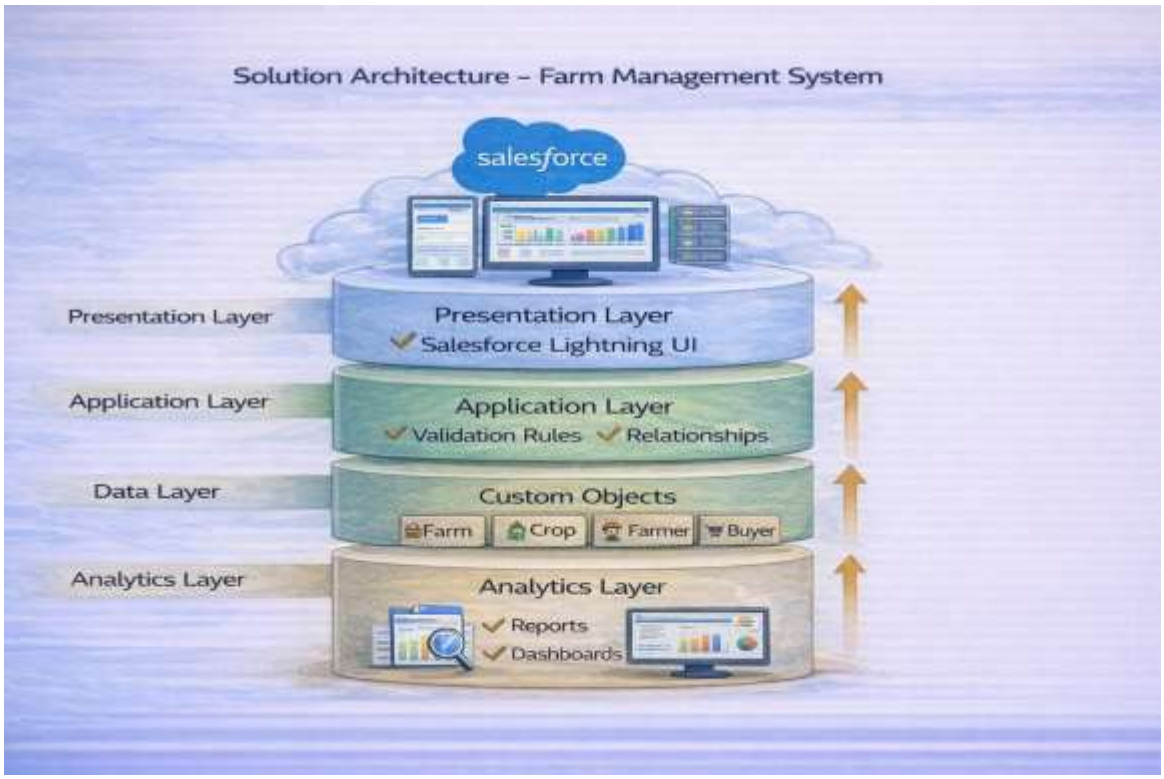
3.1 Problem-Solution

Fit The Farm Management System (FMS) is designed to bridge the gap between traditional manual farming records and modern digital data management. By understanding the pain points of farmers and farm managers, the system provides a tailored Salesforce cloud solution that completely eliminates paper-based data loss, manual errors, and communication delays.



3.2 Proposed Solution

Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and reporting delays.
Solution	A centralized Salesforce-based Farm Management System to digitize and automate all agricultural records securely.
Target Users	Farmers, Farm Managers, Buyers, and System Administrators.
Key Benefits	Centralized information, real-time analytics, faster reporting, automated data validation, and better decision-making.



Phase 4: Project Planning Phase

4.1 Project Methodology

The Farm Management System (FMS) project follows the Agile methodology to ensure smooth and continuous development. The implementation is divided into multiple phases (Sprints) to effectively manage requirement gathering, Salesforce object setup, automation, and testing.

4.2 Sprint Plan

Sprint	Activities	Target Completion
Sprint 1	User Story, Salesforce Account Creation, and Project Planning	Week 1
Sprint 2	Objects, Tabs, Lightning App, Fields and Relationship	Week 2
Sprint 3	Profile, Role, Users, Sharing Rules, and User Adoption	Week 3
Sprint 4	Reports, Dashboard, View Report and Dashboard	Week 4

4.3 Task Allocation

Team Member	Role	Assigned Tasks (Based on Workspace)
S.Balamurugan	Leader	Project Tracking, User Story, Salesforce Account Creation, User Adoption
M .Dharshini	Member	Objects, Tabs, Lightning App
V . Dhanush	Member	Fields and Relationship
P . kaviyarasan	Member	Profile, Role, Users, Sharing Rules
R . Poornima	Member	Reports, Dashboard, View Report and Dashboard

PHASE 5: PROJECT DEVELOPMENT PHASE

Project Description

The Farm Management System (FMS) is a cloud-based application developed using the Salesforce platform to digitize and streamline agricultural operations. It replaces traditional, manual paper-based record-keeping with a secure, centralized digital system. The project focuses on managing core entities such as Farms, Crops, Farmers, and Buyers. It enables users to accurately track crop cycles, manage agricultural sales, and generate real-time analytics to improve efficiency and support data-driven decision-making.

Project Milestones

The development of the Farm Management System is divided into several key milestones to ensure smooth implementation:

- **Milestone 1: Initial Setup** (User Story finalization, Salesforce Account Creation, and Planning).
- **Milestone 2: Data Modeling** (Creation of Custom Objects like Farm, Crop, etc., along with Fields and Relationships).
- **Milestone 3: User Interface (UI) Development** (Configuration of Custom Tabs and the FMS Lightning App).
- **Milestone 4: Security & Access Control** (Setup of Profiles, Roles, Users, and Sharing Rules to ensure data privacy).
- **Milestone 5: Analytics & Finalization** (Creation of real-time Reports, interactive Dashboards, and User Adoption strategies).

What is Salesforce?

Salesforce is a leading cloud-based Customer Relationship Management (CRM) platform that provides Software-as-a-Service (SaaS) and Platform-as-a-Service (PaaS) solutions. It allows businesses and developers to build custom applications quickly and securely without worrying about managing complex IT infrastructure. In this project, the Salesforce Developer Edition is used to store agricultural data safely, design a user-friendly interface using Salesforce Lightning, implement role-based data security, and generate automated dashboards.

Salesforce Account Creation

Activity1

Creating a Salesforce Developer Org

Go to https://developer.salesforce.com/signup

On the sign up form, enter the following details :

First name & Last name

Email

Role : Developer

Company : College Name

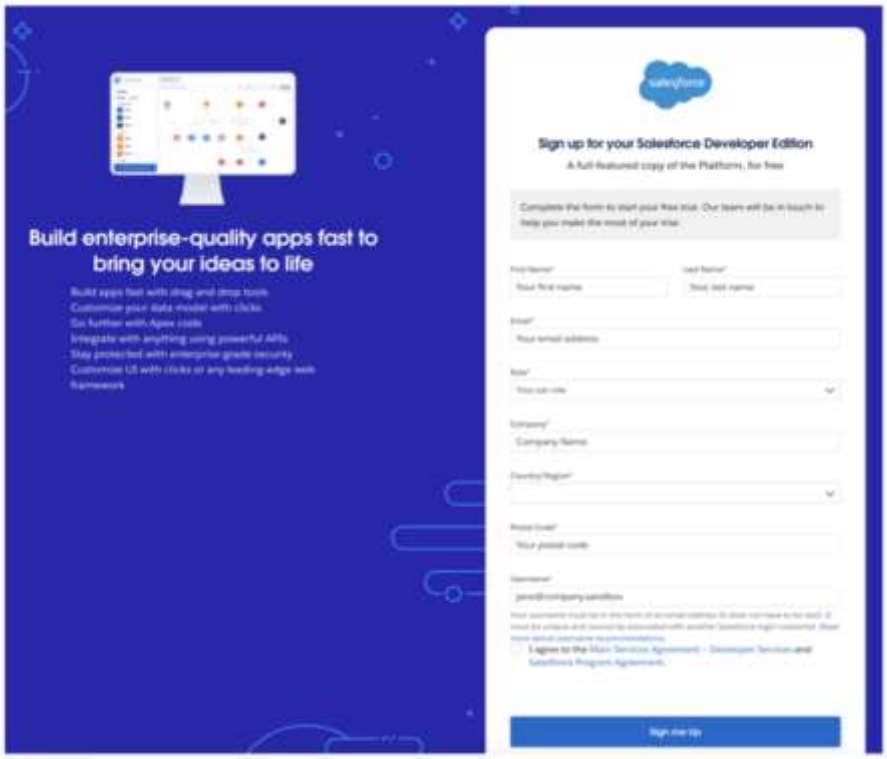
County : India

Postal Code : pin code

Username : should be a combination of your name and company

This need not be an actual email id, you can give anything in the format : username@organization.com

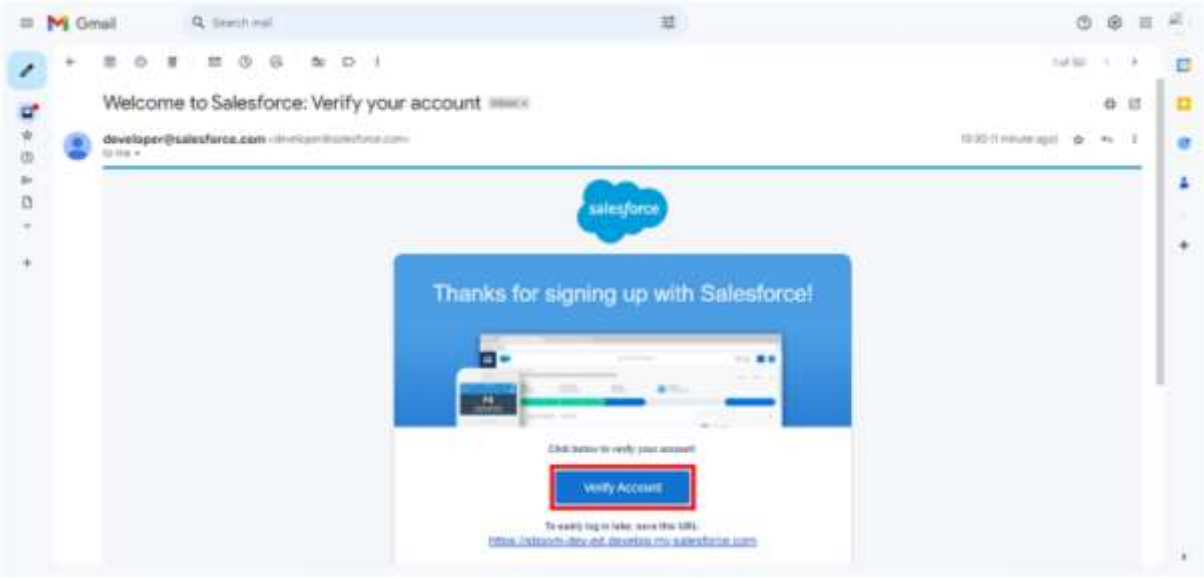
Click on sign me up after filling these.



Activity 2 :

Account Activation

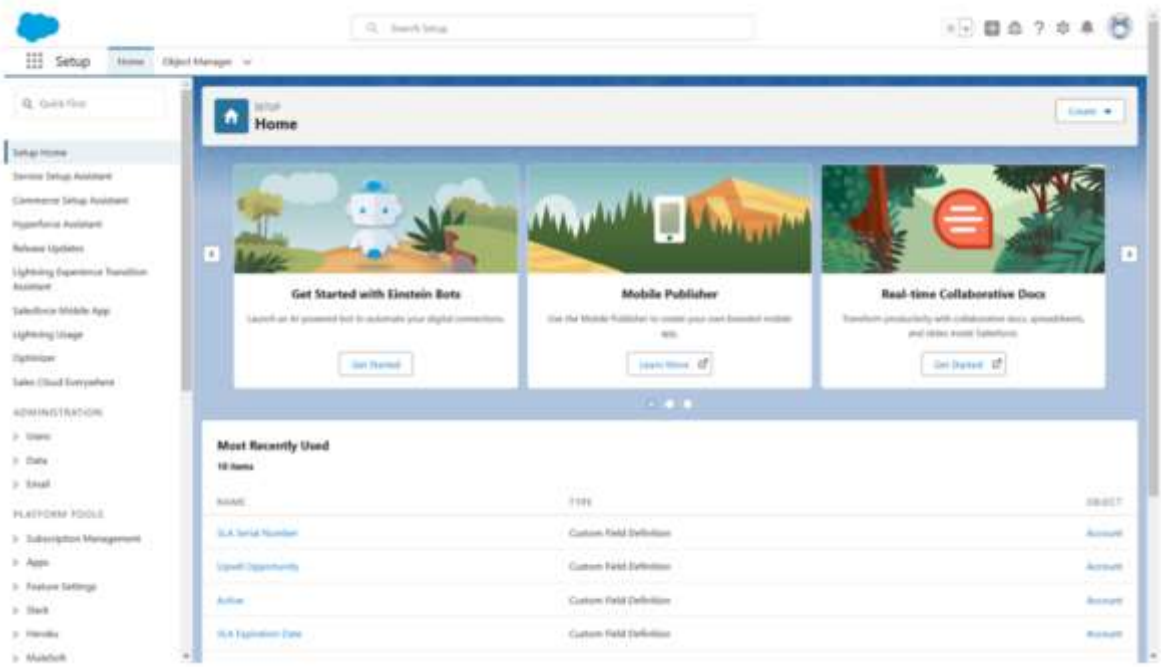
Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.



2. Click on Verify Account

3.Give a password and answer a security question and click on change password.

4.Then you will redirect to your salesforce setup page.



Objects

Activity 1 : Create a custom object for Farm

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

1.Farm

2.Crop

3.Farmer

4.Buyer

Activity 2:Creation of Crop object

Click on the gear icon and then select Setup.

Click on the object manager tab just beside the home tab.

After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object. a. On the Custom Object Definition page, create the object as follows:

b. Label: Crop

c. Plural Label: Crops

d. Record Name: Crop Name

e. Select the data type as "Text".

f. Check the Allow Reports checkbox

g. Check the Allow Search checkbox

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout

i. Click Save.

Activity 3 : Creation of Farmer object

Click on the gear icon and then select Setup.

Click on the object manager tab just beside the home tab.

After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows:

b. Label: Farmer

c. Plural Label: Farmers

d. Record Name: Farmer Name
e. Select the data type as "Text".

f. Check the Allow Reports checkbox.

g. Check the Allow Search checkbox.

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.

i. Click Save.

Activity 4:Creation of Buyer object

Click on the gear icon and then select Setup.

Click on the object manager tab just beside the home tab.

After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object. a. On the Custom Object Definition page, create the object as follows:

b. Label: Buyer

c. Plural Label: Buyers

d. Record Name: Buyer Name

e. Select the data type as "Text".

f. Check the Allow Reports checkbox.

g. Check the Allow Search checkbox.

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.

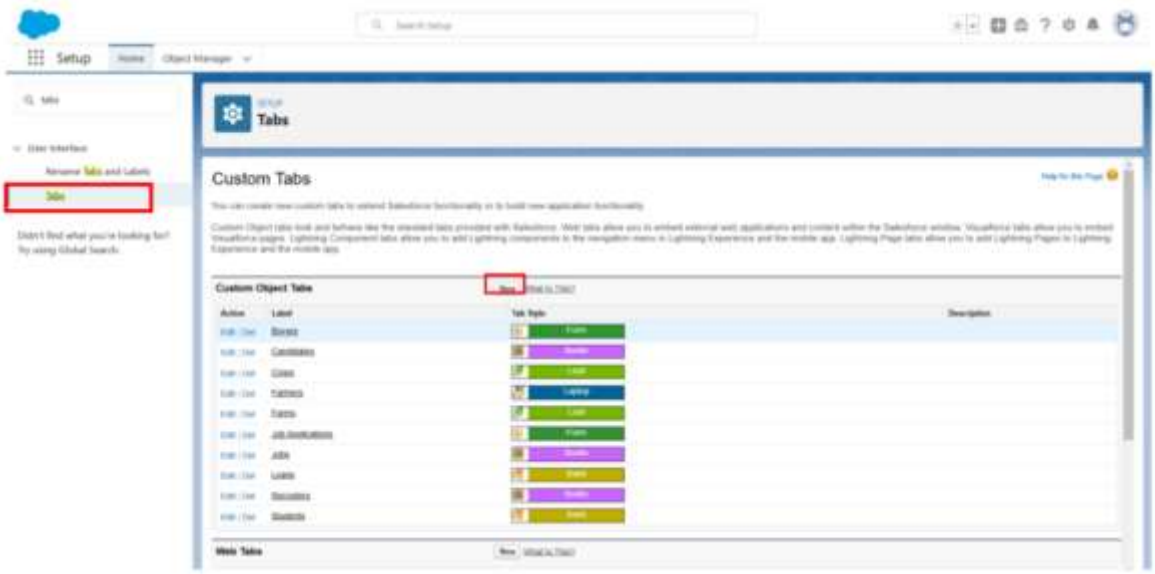
i. Click Save

Tabs

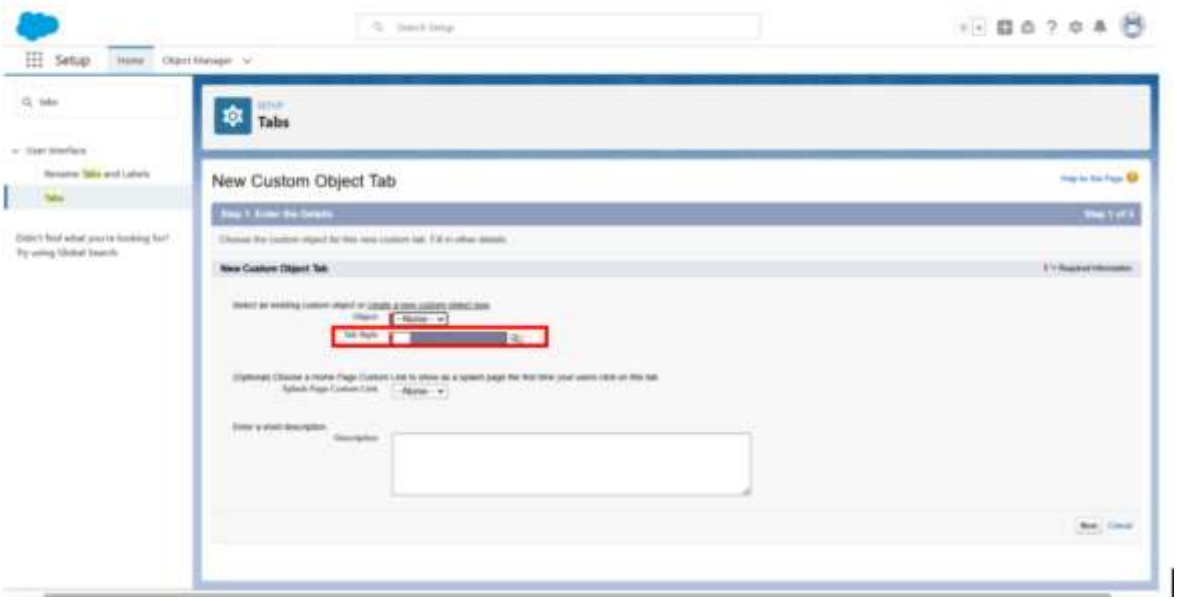
Activity 1: Creation of Farm tab

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2. Under custom object tabs, click New.



For Object, select Farm.

For Tab Style, select any icon.

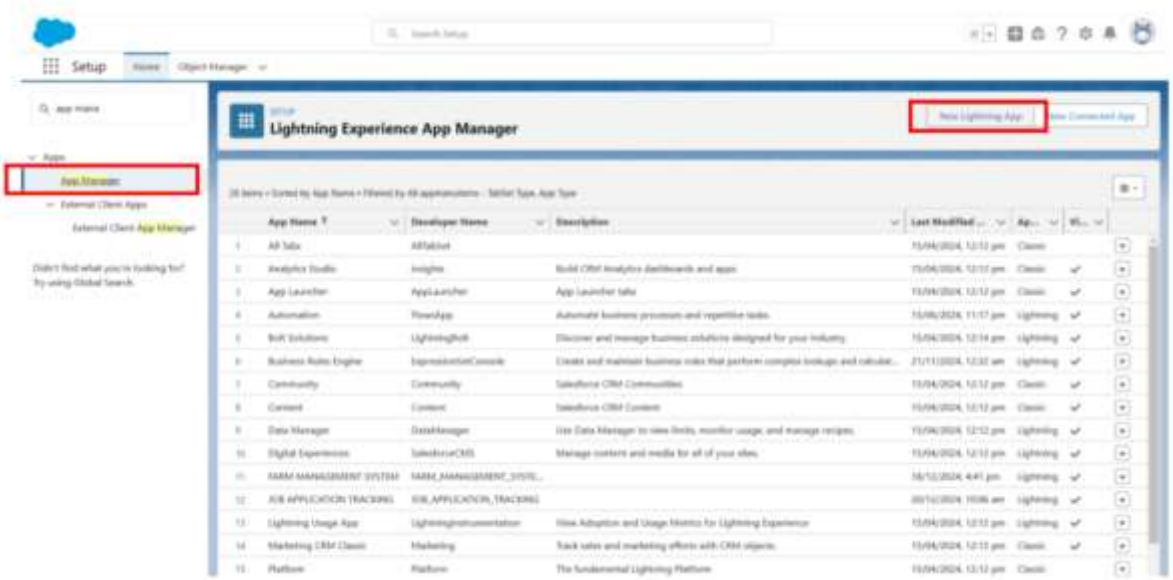
Leave all defaults as is. Click Next, Next, and Save.

Lightning App

Activity 1: Create the Farm Management System app

● From Setup, enter App Manager in the Quick Find and select App Manager.

● Click New Lightning App. Farm Management System as the App Name, then click Next



Under App Options, leave the default selections and click Next.

● Under Utility Items, leave as is and click Next.

● From Available Items, select Farm, Crop, Farmer ,Buyer Reports ,and Dashboards and move them to Selected Items. Click Next.

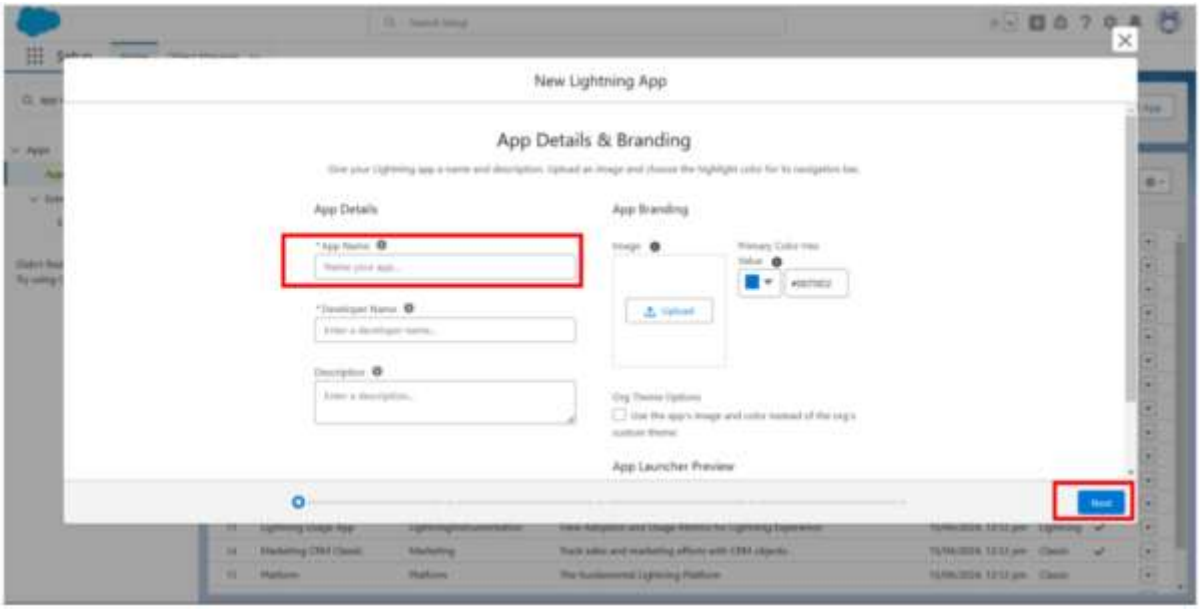
● From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.

● To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

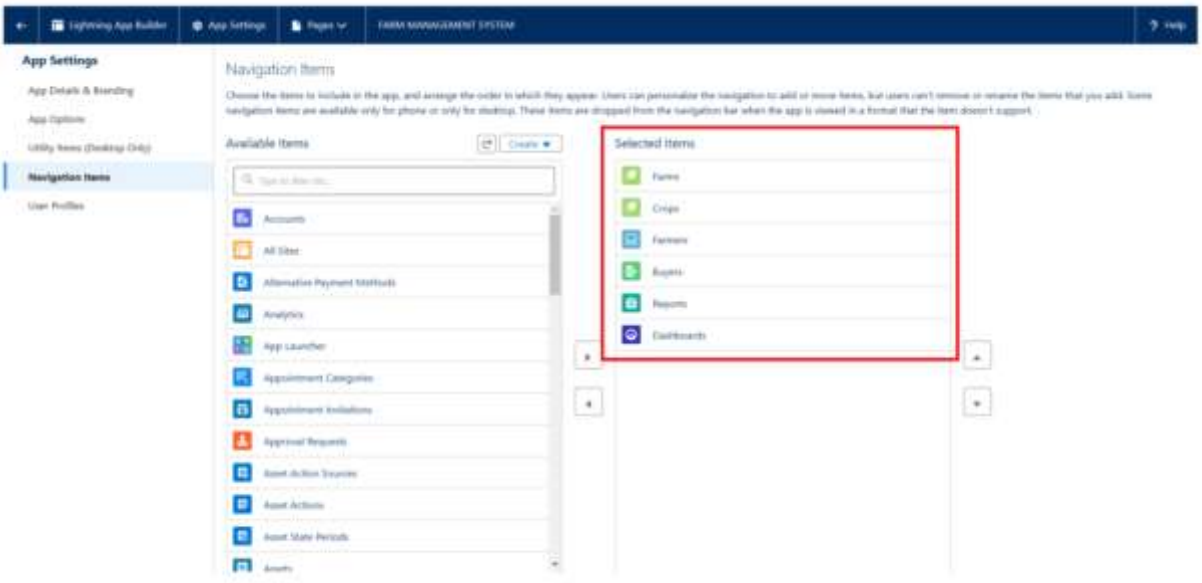
Note :

App Launcher-Displays available apps.

App Name-Displays the current selected app.



Navigation menu-Displays the tabs available inside the app.



Fields and Relationship

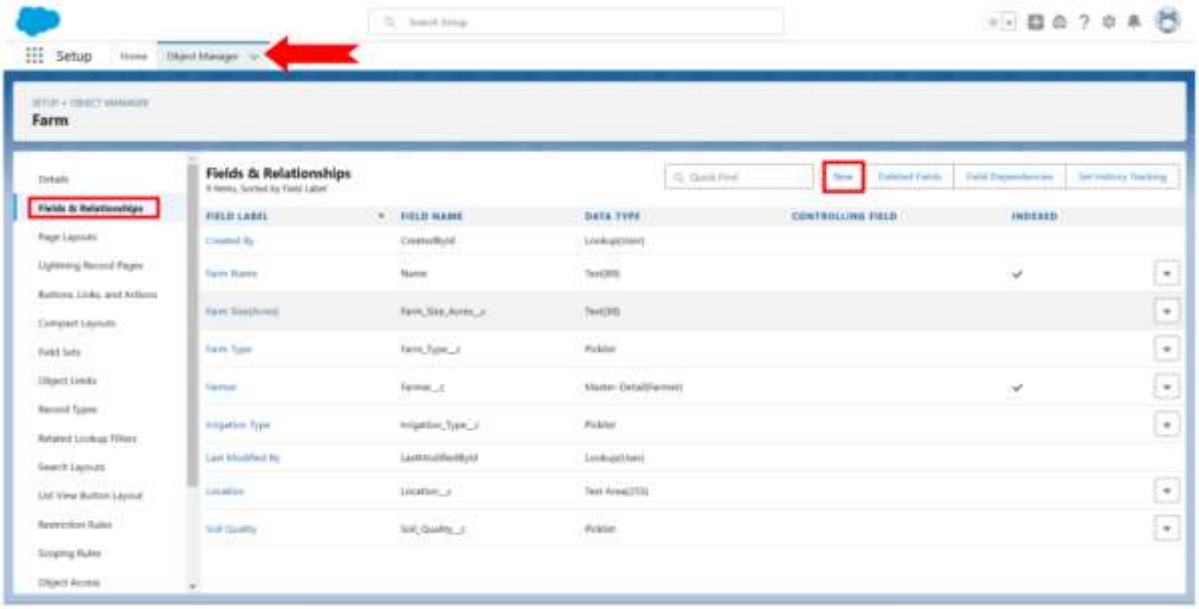
Activity 1:Creation of fields for the Farm objects

click the gear icon and select Setup. This launches Setup in a new tab.

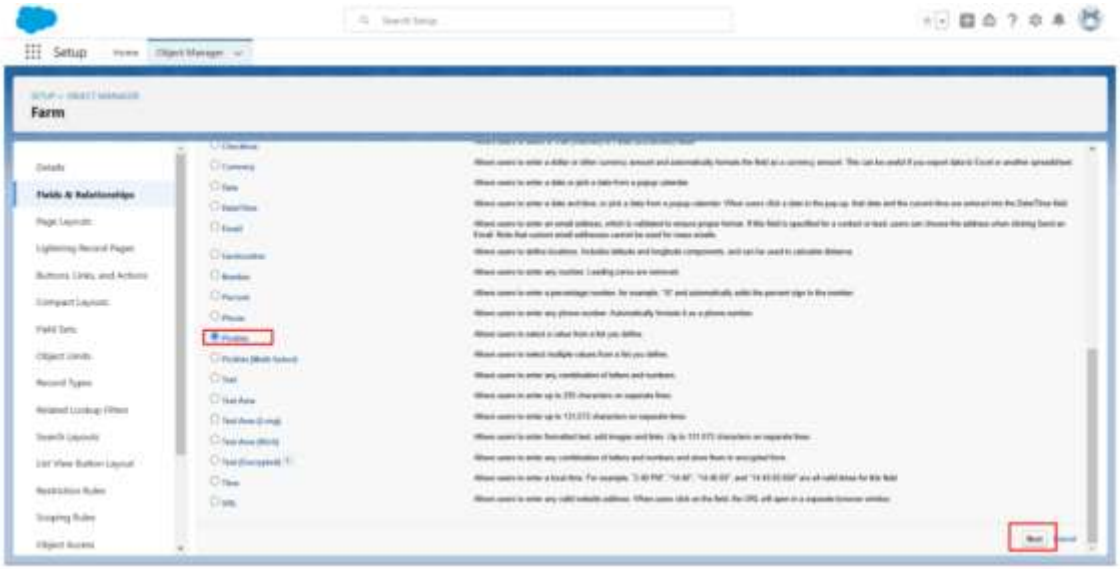
Click the Object Manager tab next to Home.

Select Farm.

Select Fields & Relationships from the left navigation, and click New

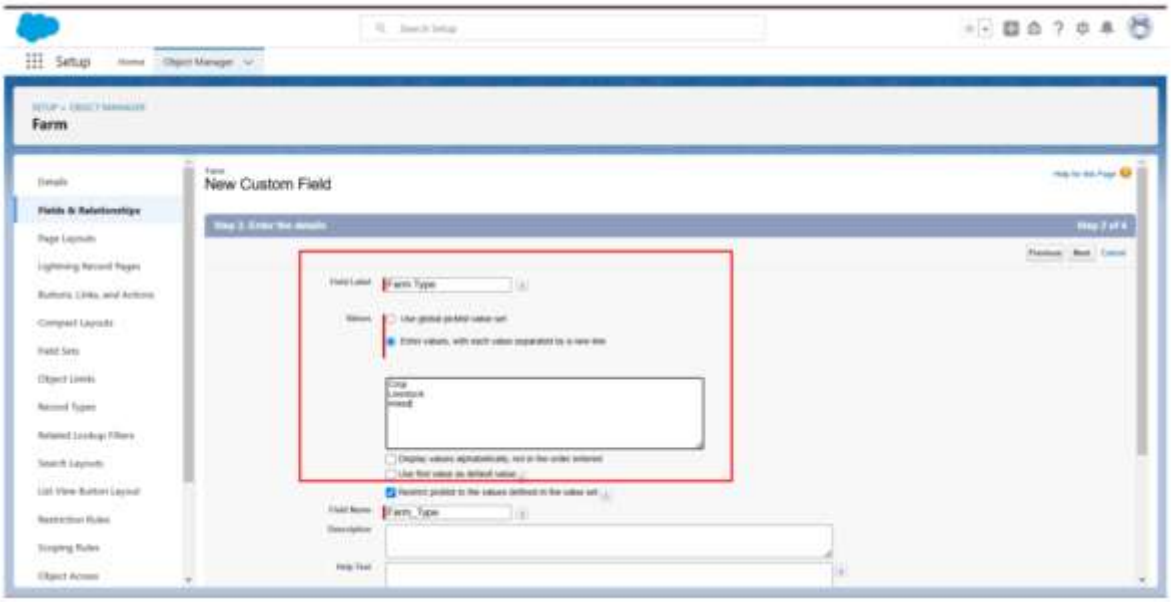


5.Choose Data type as a picklist ,Click next



6.Enter Field Label (Farm Type),and Enter Values as (Crop,Livestock,Mixed)

7.Next, Next Click Save.



Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)

2.Again we Create Picklist field,Field Label (Irrigation type),

Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)

3.Choose data type as a text,enter field label(Location)

4.Choose data type as text,enter field label (size)

5.Create a Master Detail relationShip with Farmer Object.

Activity 2: Creation of fields for the Crop objects

1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save.

2.Select Data type as date ,Enter Field label (Planting Date),then save .

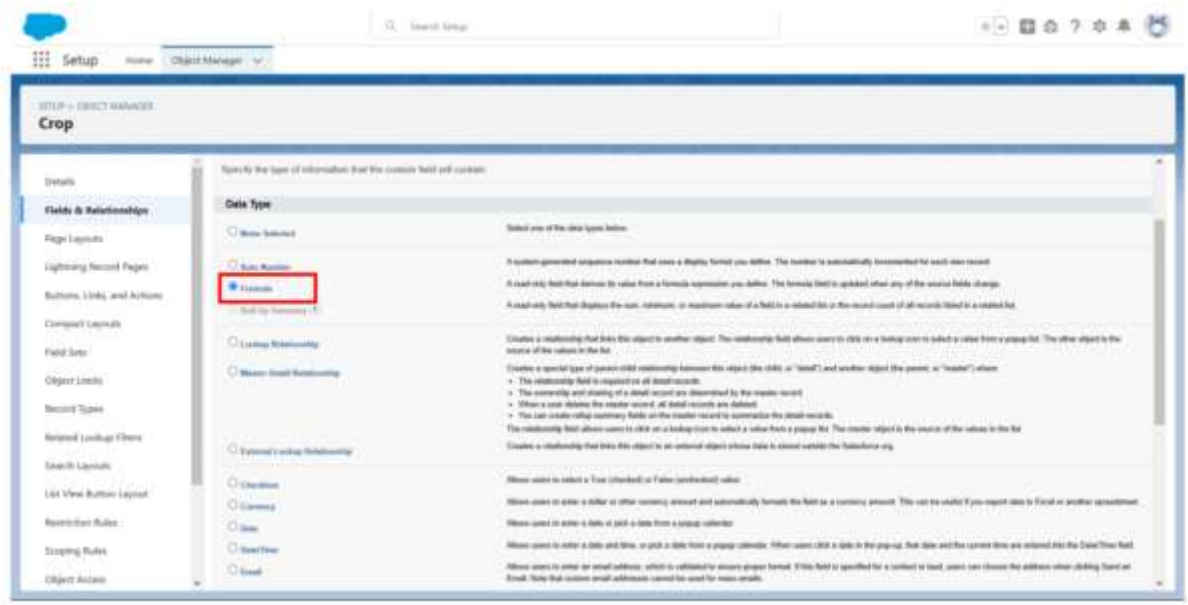
3.Again we select data type as a date,Enter field label (Harvest date),save

4.Next We Create Lookup relationships on crop objects related to Farm Objects.

5.We select data type as currency,Enter field Label (Actual Yield), Next Save it.

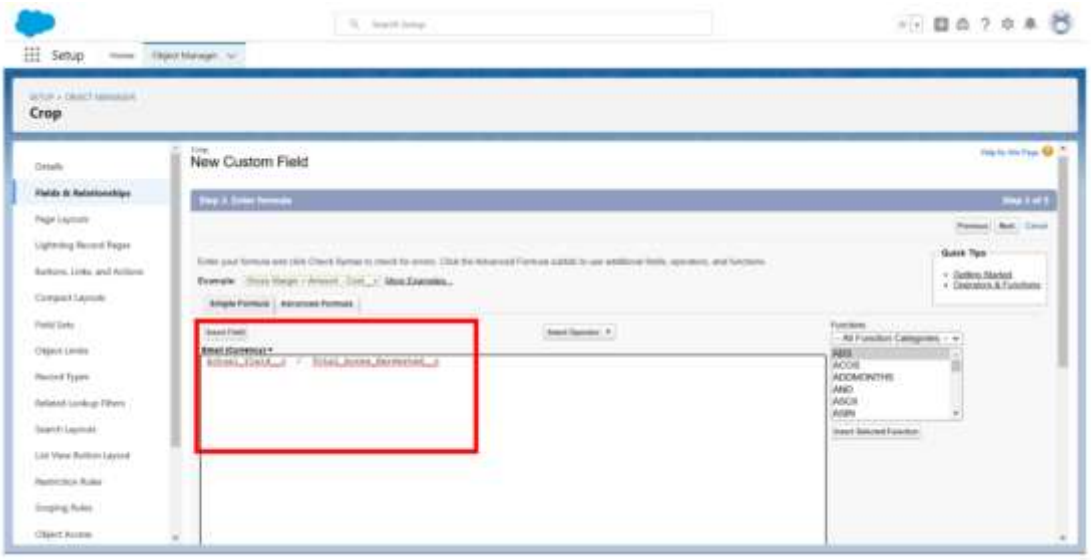
6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.

7.Now We create a Formula Field For That Select data Type As a Formula,Click Next.



8.Enter Field Label (Expected Yield per Acre),Select formula return type as Currency,click Next.

9.Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.



Activity 3: Creation of fields for the Farmer objects

1.Select Data type Phone,Enter field label(Contact Number),next Save.

2.Select Data type Picklist,Enter Field Label(State),Enter values (Telangana,Andhra Pradesh,Tamilnadu),save.

3.Choose Data type Picklist,Enter Field Label (District),Enter Values of district,next save.

4.Again select Data type as Picklist,Enter Field Label (Village),Enter Values of villages,next save.

Creation of field dependency :

1.Now create Field Dependency For District field,Open Object Manager,Choose Farmer object ,select Fields&Relationships,On Right side We have Field Dependency option,select that option.

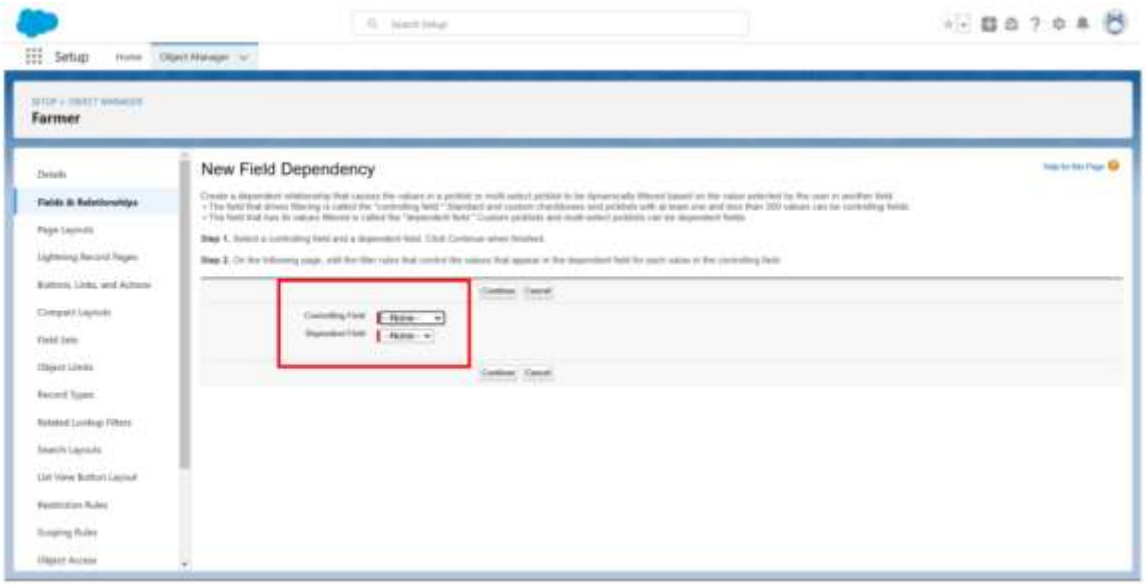
2.Click new, Select Controlling Field as State,Dependent Field As District,Click on Continue.

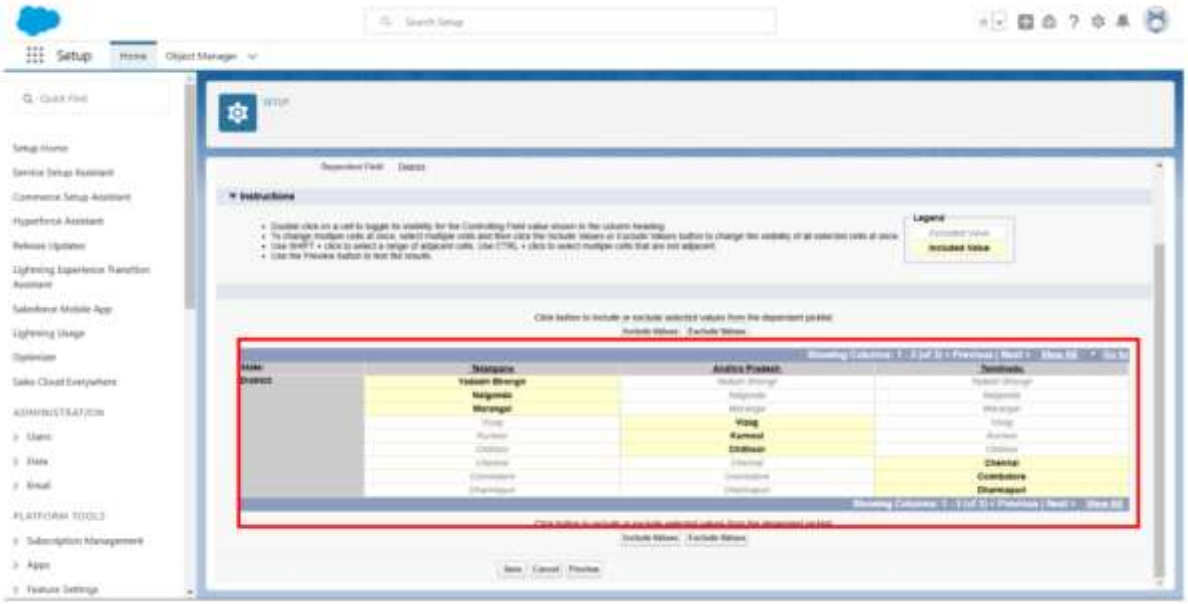
3.Select the appropriate Value in each column by double-clicking them.

4. For Ex. Telangana - Yadadri Bhongir.

5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.

6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.





7. Click Close to close the preview window, Click Save.

Activity 4: Creation of fields for the Buyer objects

1.Select Data type as a Phone,Enter a Field Label(Contact),Save.

2.Select Data type as an Email,Enter Field Label(Email), Next save.

3.Select Data type Text area,Enter Field Label(Address),Save it.

4.Select Data type Picklist, Field Label(Buyer Type),Enter values (Wholesale,Retail),Click Save.

Profile

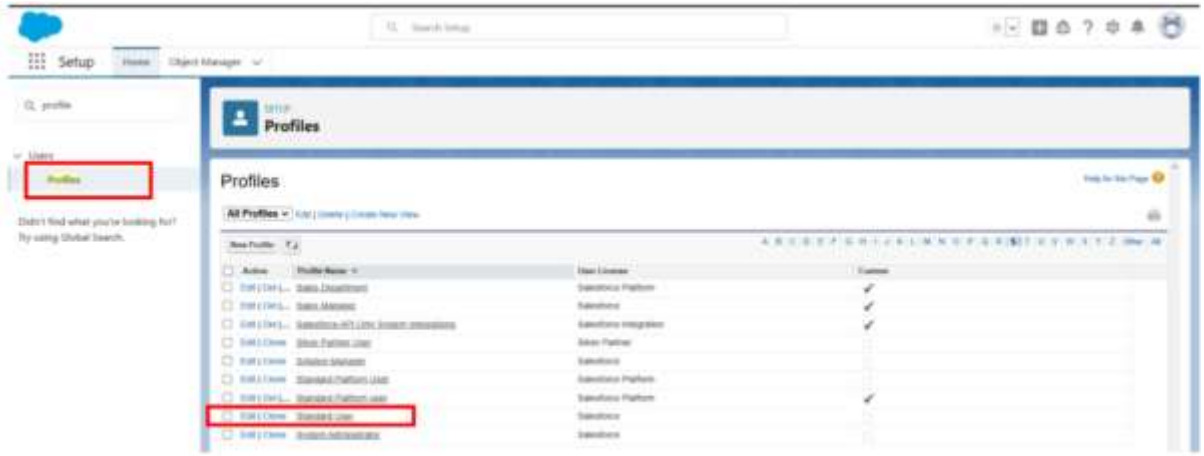
Activity 1: Create a custom profile

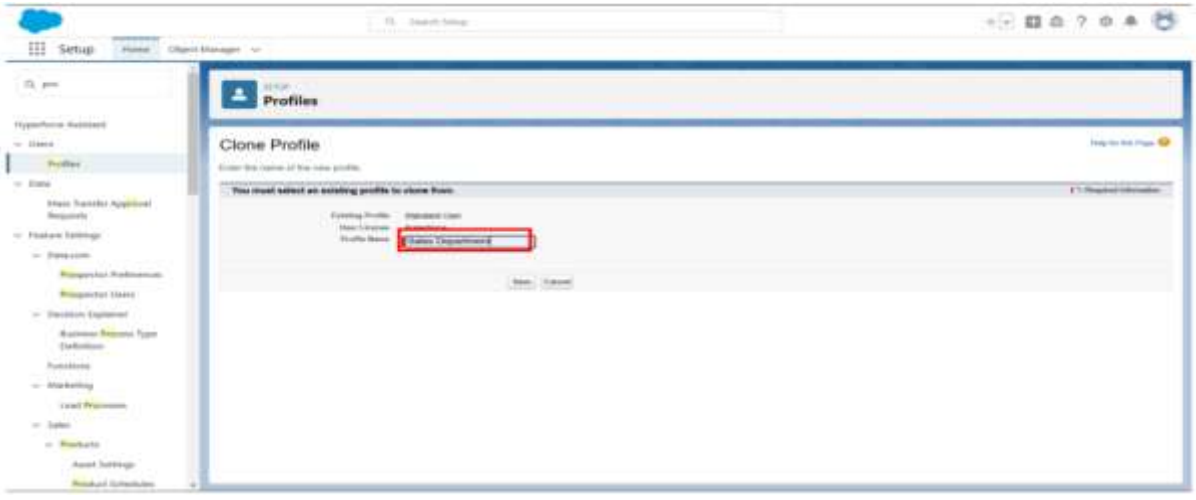
From setup , enter profiles in Quick Find box

Select profiles (Standard user).

Click clone.

For Profile, enter the Sales Department.





Activity 2: Create a custom profile-2

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box

7. Select profiles (Standard user).

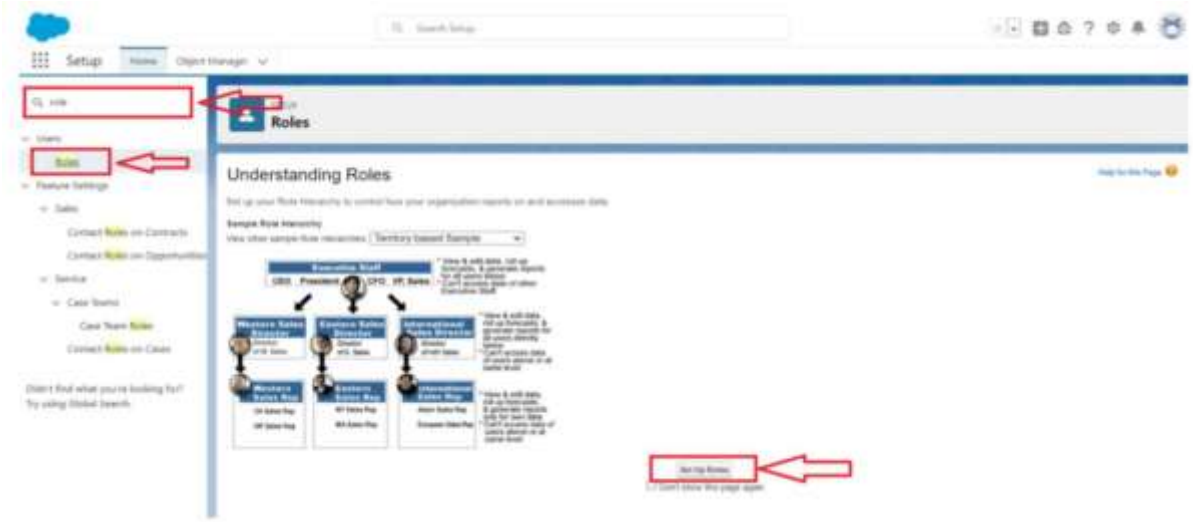
8. Click clone**.**

Role

Activity-1: Creation of role

From the Quick find box search for the role and click on the roles option

select the set up roles option



3. Below the CEO click on add role and enter the label name as a ” Farm Manager ” and the role name will be Automatically populated and click on save

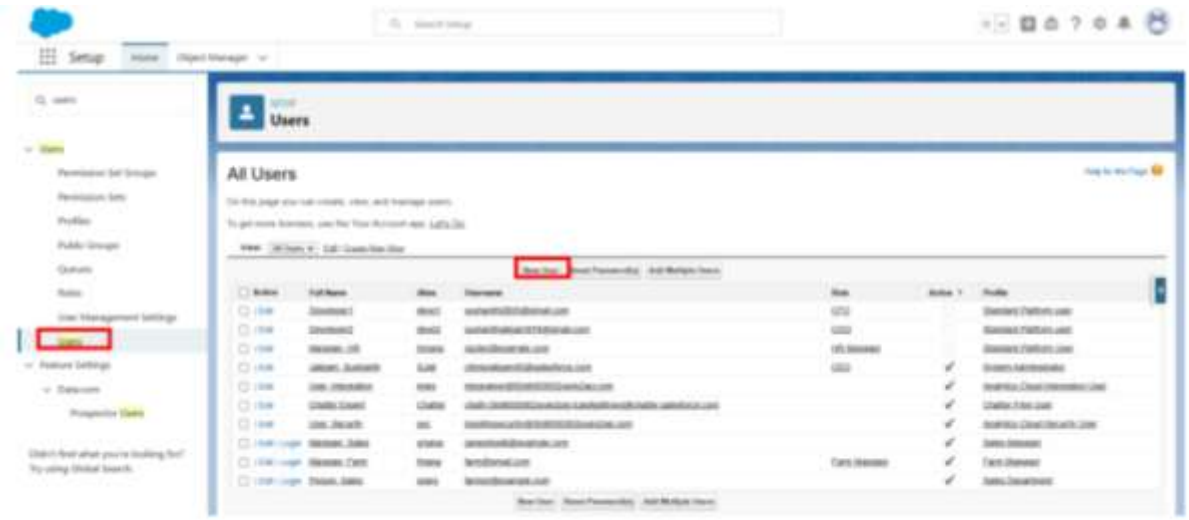


Users

Activity 1: To Create a user

From Setup, enter Users in the Quick Find box, then select Users.

Click New User.



Enter First name as Farm and last name as Manager.

Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.

Select role Farm Manager.

Select user License as Standard Platform User.

Select profile(Farm Manager).

Click save.

Activity 2: Create a user with a username as Sales Department

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Sharing Rules

Activity 1:Create a sharing rule

1) Go to Sharing Settings, which can be found under the Quick Find section.

2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.

3) Add the label of the sharing rule you want to make.

4) Select your rule type based on the criteria.

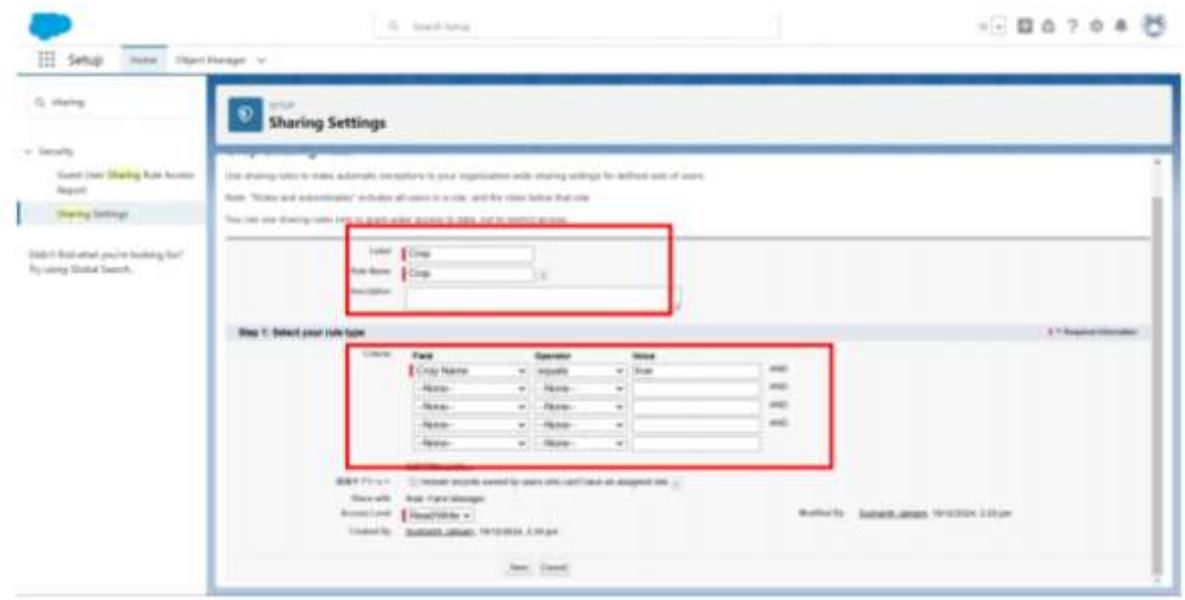
5) Select the field and immediately check the field from the Crop object.

6) Select the crop as equal and the value is True.

7) And in selecting the users to share with the section select roles and in that select Farm Manager.

8) And in the section of select the level of access for the users give the access Read/Write.

9) And save the rule.



Activity 2:Create a Sharing Rule to Share the records of Buyer

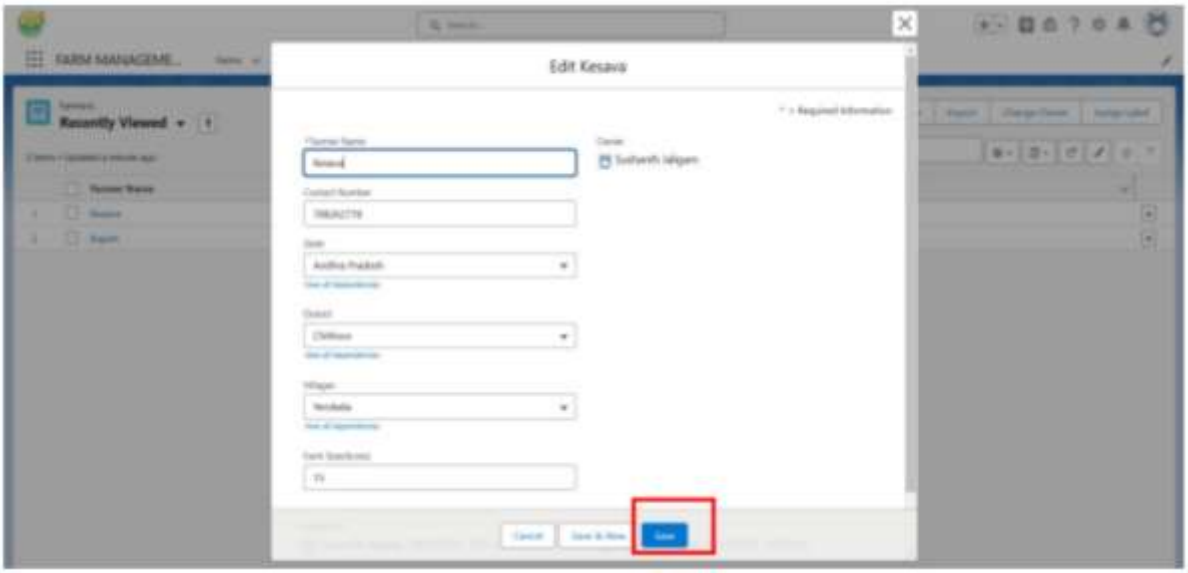
Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity

User Adoption

Activity 1:Create a Record(Farmer)

- Click on App Launcher on the left side of the screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.



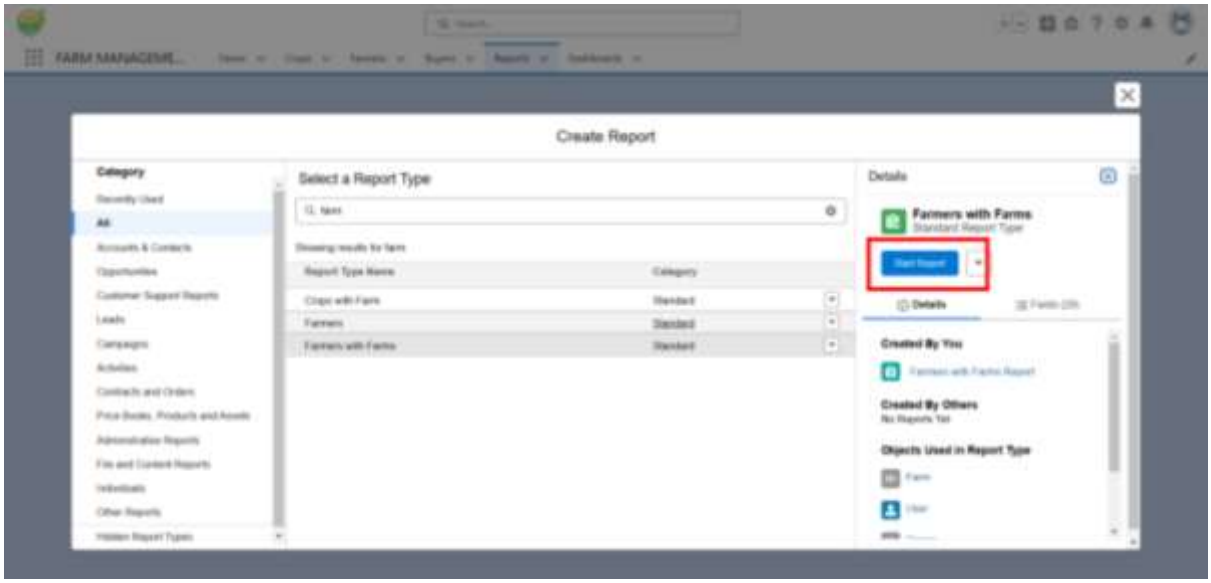
Reports

Activity 1:Create a report

Create a report that displays the rating of the account and which has type and account name.

1) Click on app launcher search for reports.

2) Click on the new report and select the category as Farmer with Farms.



3) In the details section select the option start report.

4) In Outline Click on Column Select Which Field you want to display in the Report.

5) Click Group,Select Which Fields You Want to Group.

6) Save the report by giving the label name(Farmer with Farms) and save the folder as a public folder and save the report.

Activity 2:Create a Report using the Objects Crops, Buyers.

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

Dashboard

Activity 1 : Create a Dashboard

Click the Dashboard tab.

Click New Dashboard.

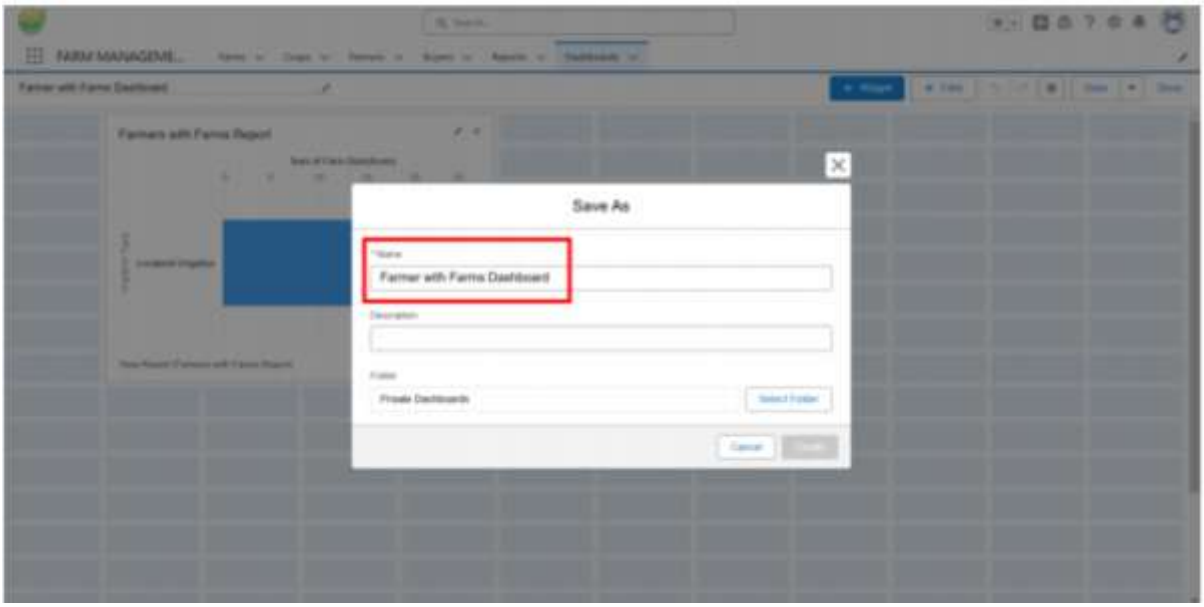
Name the dashboard Farmers with farms and click Create.

Click +Component.

Select the Farmers with Farms and click Select.

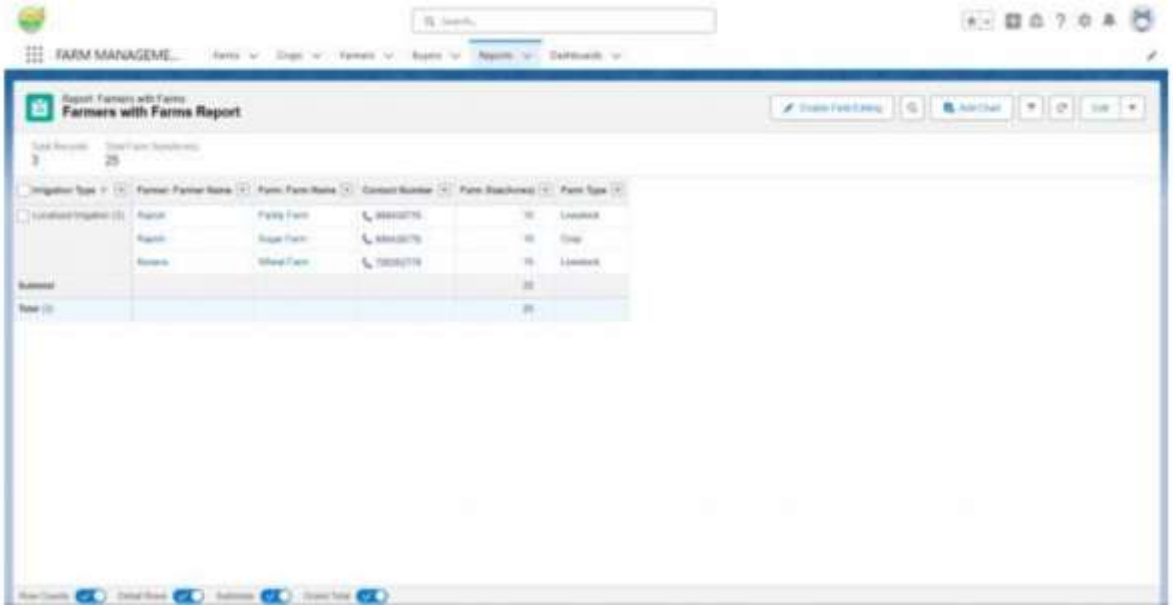
Select the Gauge chart and click Add.

Click Save and then Done.



View Report and Dashboard

- Click on App Launcher on the left side of the screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



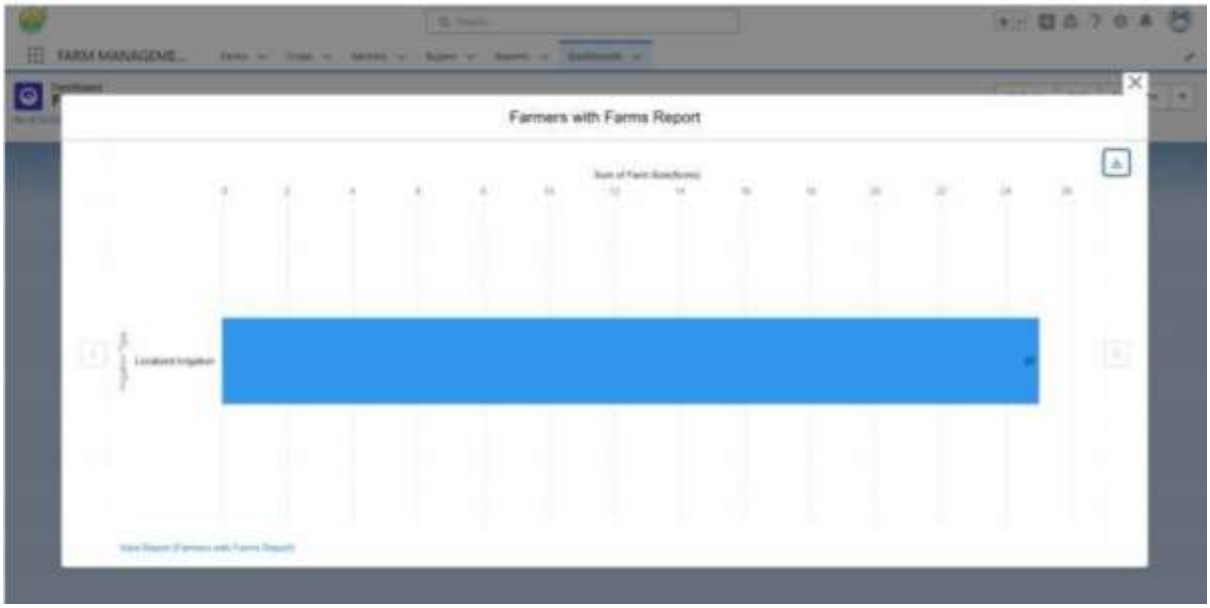
Activity 2: Dashboard

• Click on App Launcher on the left side of the screen.

• Search Farm Management system & click on it.

• Click on Dashboard Tab.

• Click Farmers With farms & see records.



PHASE 6: SOFTWARE TESTING PHASE

6.1 Testing Strategy

The testing phase for the Farm Management System (FMS) was conducted to ensure all functional and non-functional requirements are met. The testing process included Unit Testing (testing individual Salesforce objects and fields), Functional Testing (validating workflows and rules), and User Acceptance Testing (UAT) to confirm the system aligns with the primary user story.

6.2 Test Environment

- Platform: Salesforce Developer Edition
- UI Environment: Salesforce Lightning Experience
- Tested By: FMS Project Team (Sabari, Santhosh, Deepika, Palani)

6.3 Detailed Test Cases

Test Case ID	Module	Test Scenario	Steps to Execute	Expected Result	Actual Result	Status
TC-01	App & Tabs	Verify FMS Lightning App accessibility	1. Login to Salesforce 2. Open App Launcher 3. Select FMS App	FMS App should open with Home, Farm, Crop, Farmer, Buyer tabs	App and tabs loaded successfully	Pass
TC-02	Object (Farm)	Verify Farm Record Creation	1. Go to Farm tab 2. Click New	New Farm record should be created and visible in list view	Record saved and displayed correctly	Pass

			3. Enter Farm details 4. Save			
TC-03	Object (Crop)	Verify Crop-Farm Relationship	1. Go to Crop tab 2. Create New Crop 3. Link to a Farm record	Crop must be successfully associated with the selected Farm	Master-Detail / Lookup link works correctly	Pass
TC-04	Validation	Verify Negative Yield Validation	1. Open Crop record 2. Enter negative number in Yield field 3. Save	System must show an error: "Yield cannot be negative"	Error message displayed; record not saved	Pass
TC-05	Security	Verify User Role Restrictions	1. Login as Standard User	User should receive an "Insufficient	Access denied message shown	Pass

			2. Attempt to delete a Farm record	Privileges" error		
TC-06	Analytics	Verify Dashboard Real-time Update	1. Add a new Farm/Crop record 2. Go to Dashboards tab 3. Refresh Dashboard	Total counts and charts should reflect the newly added data	Dashboard metrics updated instantly	Pass

6.4 Bug Tracking & Resolution

During the testing phase, minor UI and validation rule errors were identified and resolved immediately using the Salesforce Object Manager and Setup menu. The final build is stable, secure, and ready for deployment.

7.1 Documentation Overview

The Project Documentation Phase involves compiling all the artifacts, diagrams, and executed steps into a comprehensive final report. This document serves as a complete guide to the Farm Management System (FMS), detailing every stage of the project lifecycle from initial brainstorming to final testing.

7.2 Documented Artifacts The following components have been successfully documented as part of the project report:

- Requirement & Design: Empathy Map, Customer Journey Map, Data Flow Diagram, and Entity-Relationship (ER) Diagram.
- Project Planning: Agile sprint plans, technology stack selection, and team task allocation (Sabari, Santhosh, Deepika, Palani).
- Development & Testing: Step-by-step Salesforce implementation records, workspace activity completion, and comprehensive test cases.

7.3 Final Submission Links As a part of the documentation and final project rollout, the project files and visual demonstrations have been recorded .

Phase 8: Conclusion Phase

8.1 Project Summary

The Farm Management System (FMS) successfully achieved its goal of digitizing agricultural record-keeping. By migrating from manual paper-based processes to a secure Salesforce cloud platform, the system ensures accurate data entry, role-based access, and real-time reporting. It effectively addresses the pain points of farmers, farm managers, and buyers by providing a centralized and automated solution.

8.2 Future Scope

- Mobile Integration: Enhancing the FMS to be fully functional on a dedicated mobile app for farmers to use directly from the field.
- AI-driven Analytics: Implementing predictive analytics to forecast crop yield and market prices based on historical data.
- Automated Notifications: Setting up automated SMS or email alerts for buyers when a specific crop is ready for harvest and sale.